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Summary of problem

Voltara, a vehicle manufacturing company, must determine the most viable option to secure 50,000 electric control units (ECU) annually over a 10-year project lifetime. There are two alternative options under consideration: Alternative A, which is establishing in-house production, and Alternative B, which is outsourcing from overseas suppliers, which will be subject to a high tariff. These options will be subject to evaluation of their net present values (NPV), internal rate of return (IRR), and overall cost-effectiveness given a 15% minimum acceptable rate of return (MARR) and a 21% corporate tax rate.

Methods and Model

I used after-tax cash flow (ATCF) analysis, NPV, and IRR as the key metrics for decision-making. The analysis was performed using Excel, where each year's depreciation, taxes, BTCF, ATCF, and present worth were calculated.

Key information used:

- ECUs Needed: 50,000 units/year
- Project Lifetime: 10 years
- Value: $90 per unit
- MARR: 15%
- Tax Rate: 21%
- Alternative A:
    - Initial Investment: $8,000,000
    - Annual Operating Costs (Excluding Depreciation): $2,500,000 per year
    - Salvage Value: $500,000
    - Useful Life: 10 years
- Alternative B:
    - Initial Investment: $200,000
    - Straight Line Depreciation: 5 years
    - Annual Cost per ECU (Before Tariffs): $33.30 per unit
    - Import Tariff: 150%
    - Annual Shipping and Handling Costs: $300,000 per year

Key equations used:

- Tax = (Revenue - Operating Costs - Depreciation) * Tax Rate
- ATCF = BTCF - Taxes
- NPV = $\sum$ (ATCF / $(1 + MARR)^t$) - Initial Investment
- IRR = Discount rate at which NPV = 0

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Given:			Calculations												
ECUs needed	50,000	per year		Alternative A: Establishing In-House Production										MARR	
Project lifetime	10	years	Years	Initial Investment	Salvage Value	Revenue	Operating Costs	MACRS	Depriciation	BTCF	Taxes	ATCF	Net Cash Flow	PW	NPV
Tax rate	21%		0	($8,000,000)						($8,000,000)		($8,000,000)	($8,000,000)	($8,000,000)	$1,080,593
MARR	15%		1			$4,500,000	$2,500,000	14.29%	$1,143,200	$856,800	$179,928	$676,872	$1,820,072	$1,582,671	
Value	$90	per ECU	2			$4,500,000	$2,500,000	24.49%	$1,959,200	$40,800	$8,568	$32,232	$1,991,432	$1,505,809	
Alternative A:			3			$4,500,000	$2,500,000	17.49%	$1,399,200	$600,800	$126,168	$474,632	$1,873,832	$1,232,075	
Initial Investment	$8,000,000		4			$4,500,000	$2,500,000	12.49%	$999,200	$1,000,800	$210,168	$790,632	$1,789,832	$1,023,342	
Annual Operating Costs (Excluding Depreciation)	$2,500,000	per year	5			$4,500,000	$2,500,000	8.93%	$714,400	$1,285,600	$269,976	$1,015,624	$1,730,024	$860,128	
Salvage Value	$500,000		6			$4,500,000	$2,500,000	8.92%	$713,600	$1,286,400	$270,144	$1,016,256	$1,729,856	$747,864	
Useful Life	10	years	7			$4,500,000	$2,500,000	8.93%	$714,400	$1,285,600	$269,976	$1,015,624	$1,730,024	$650,380	
			8			$4,500,000	$2,500,000	4.46%	$356,800	$1,643,200	$345,072	$1,298,128	$1,654,928	$540,999	
			9			$4,500,000	$2,500,000			$2,000,000	$420,000	$1,580,000	$1,580,000	$449,135	
Alternative B:			10		$500,000	$4,500,000	$2,500,000			$2,500,000	$525,000	$1,975,000	$1,975,000	$488,190	
Initial Investment	$200,000											3%			
Straight Line Depriciation	5	years										IRR			
Annual Cost per ECU (Before Tariffs)	$33.30	per unit													
Import Tariff	150%														
Annual Shipping and Handling Costs	$300,000	per year													

	Alternative B: Overseas Sourcing (with Tariffs)										MARR		
Years	Initial Investment	Depriciation	Revenue	Purchase Cost	S&H Cost	Total Cost	BTCF	Tax	ATCF	Net Cash Flow	PW	NPV	Incremental Cash Flows
0	($200,000)						($200,000)		($200,000)	($200,000)	($200,000)	($23,161)	-$7,800,000
1		$40,000	$4,500,000	$4,162,500	$300,000	$4,462,500	-$2,500	-$525	-$1,975	$38,025	$33,065		$678,847
2		$40,000	$4,500,000	$4,162,500	$300,000	$4,462,500	-$2,500	-$525	-$1,975	$38,025	$28,752		$34,207
3		$40,000	$4,500,000	$4,162,500	$300,000	$4,462,500	-$2,500	-$525	-$1,975	$38,025	$25,002		$476,607
4		$40,000	$4,500,000	$4,162,500	$300,000	$4,462,500	-$2,500	-$525	-$1,975	$38,025	$21,741		$792,607
5		$40,000	$4,500,000	$4,162,500	$300,000	$4,462,500	-$2,500	-$525	-$1,975	$38,025	$18,905		$1,017,599
6			$4,500,000	$4,162,500	$300,000	$4,462,500	$37,500	$7,875	$29,625	$29,625	$12,808		$986,631
7			$4,500,000	$4,162,500	$300,000	$4,462,500	$37,500	$7,875	$29,625	$29,625	$11,137		$985,999
8			$4,500,000	$4,162,500	$300,000	$4,462,500	$37,500	$7,875	$29,625	$29,625	$9,684		$1,268,503
9			$4,500,000	$4,162,500	$300,000	$4,462,500	$37,500	$7,875	$29,625	$29,625	$8,421		$1,550,375
10			$4,500,000	$4,162,500	$300,000	$4,462,500	$37,500	$7,875	$29,625	$29,625	$7,323		$1,945,375
									-4%				3%
									IRR				IRR

Figure 3: Detailed spreadsheet of all calculations for both alternatives

Questions

1. Develop a detailed spreadsheet model to calculate the before- and after-tax cash flows for each alternative over a 10-year period.
    a. See Figure 3 above.
2. Calculate the Internal Rate of Return (IRR) for each sourcing option and the incremental IRR, based on the after-tax cash flows.
    a. Alternative A has an IRR of 3%, Alternative B has an IRR of -4%, and the incremental IRR is 3% when based on ATCF.
3. If Voltara's minimum acceptable rate of return (MARR) is 15%, which sourcing strategy should they pursue based on the incremental IRR analysis?
    a. The incremental NPV (A - B) is $1,782,047, confirming that A strongly dominates B economically. The incremental IRR is 3%, based on the difference in ATCFs between the two alternatives, reinforcing Alternative A as the superior choice.
4. Identify and briefly discuss at least two significant non-financial factors that Voltara should consider before finalizing their decision between in-house production and overseas sourcing.
    a. Voltara should consider supply chain reliability like overseas sourcing delays, tariffs, and instability, while in-house offers control and consistency; and intellectual property and quality protection like in-house safeguards proprietary designs and ensures product standards, unlike overseas suppliers with weaker enforcement.

Executive Summary

After financial modeling and comparison, Alternative A (In-House Production) is clearly the better sourcing strategy. With a positive NPV of over $1 million, an IRR above that of Alternative B, and decent tax efficiency through MACRS depreciation, it surpasses Alternative B. The overseas sourcing option incurs significant tariff and handling costs that eat into profitability. Therefore, the company should proceed with Alternative A which is establishing in-house ECU production.